# ONLINEBOOKSTORE — SQL PROJECT

MySQL Data Analysis | Prepared by Rishu Gupta | 20 Questions & Solutions

## BASIC QUESTIONS & MYSQL SOLUTIONS

### 1. Retrieve all books in the "Fiction" genre

```
SELECT *
FROM Books
WHERE Author = 'Fiction';
```

### 2. Find books published after the year 1950

```
SELECT * FROM books
WHERE Published_Year > 1950;
```

### 3. List all customers from the Canada

```
SELECT *
FROM Customers
WHERE Country = 'Canada';
```

### 4. Show orders placed in November 2023

```
SELECT *
FROM orders
WHERE Order_Date BETWEEN '2023-11-01' AND '2023-11-30';
```

### 5. Retrieve the total stock of books available

```
SELECT SUM(Stock) AS Total_Books_Available
FROM books;
```

### 6. Find the details of the most expensive book.

```
SELECT * FROM books
ORDER BY Price DESC
LIMIT 1;
```

### 7. Show customers who ordered more than one quantity of a book.

```
SELECT c.Customer_ID,
       c.Name,
       c.Email,
       c.Phone,
       c.City,
       c.Country,
       o.Quantity
FROM customers AS c
INNER JOIN orders AS o ON c.Customer_ID = o.Customer_ID
WHERE o.Quantity > 1;
```

### 8. Retrieve orders where the total amount exceeds $20.

```
SELECT *
FROM orders
WHERE Total_Amount > 20;
```

### 9. List all genres available in the Books table.

```
SELECT DISTINCT Genre FROM books;
```

### 10. Find the book with the lowest stock.

```
SELECT * FROM books
ORDER BY Stock ASC
LIMIT 1;
```

### 11. Calculate the total revenue generated from all orders.

```
SELECT SUM(Total_Amount) AS Total_Revenue
FROM orders;
```

## ADVANCED QUESTIONS — MYSQL SOLUTIONS

### 1. Retrieve the total number of books sold for each genre:

```
SELECT b.Genre, SUM(o.Quantity) AS Total_Books_Sold
FROM orders o
JOIN books b
ON o.Book_ID = b.Book_ID
GROUP BY b.Genre;
```

### 2. Find the average price of books in the "Fantasy" genre:

```
SELECT Genre, AVG(Price) AS Average_Price
FROM books
WHERE Genre = "Fantasy";
```

### 3. List customers who have placed at least 2 orders:

```
SELECT c.Name, COUNT(o.Customer_ID) AS No_Of_Order
FROM Customers AS c
INNER JOIN orders AS o
ON c.Customer_ID = o.Customer_ID
GROUP BY c. Name
HAVING No_Of_Order >=2
ORDER BY No_Of_Order DESC;
```

### 4. Find the most frequently ordered book:

```
SELECT b.Title, o.Book_ID, COUNT(o.Order_id) AS Order_Count
FROM orders o
INNER JOIN books b
ON o.Book_ID = b.Book_ID
GROUP BY o.Book_id
ORDER BY Order_Count DESC
LIMIT 1;
```

### 5. Show the top 3 most expensive books of 'Fantasy' Genre :

```
SELECT * FROM books
WHERE Genre = "Fantasy"
ORDER BY Price DESC
LIMIT 3;
```

### 6. Retrieve the total quantity of books sold by each author:

```
SELECT b.Author, SUM(o.Quantity) AS Quantity_sold
FROM orders o
JOIN books b
ON o.Book_ID = b.Book_ID
GROUP BY b.Author
ORDER BY Quantity_sold DESC;
```

### 7. List the cities where customers who spent over $30 are located:

```
SELECT DISTINCT c.City, Total_Amount
FROM orders o
JOIN customers c
ON o.Customer_ID = c.Customer_ID
WHERE o. Total_Amount > 30;
```

### 8. Find the customer who spent the most on orders:

```
SELECT c.Customer_ID, c.Name, SUM(o.Total_Amount) AS Total_Spent
FROM customers c
JOIN orders o
ON c.Customer_ID = o.Customer_ID
GROUP BY c.Customer_ID , c.Name
ORDER BY Total_Spent DESC
LIMIT 1;
```

### 9. Calculate the stock remaining after fulfilling all orders:

```
SELECT
b.Book_ID,
b.Title,
b.Stock,
COALESCE(SUM(o.Quantity), 0) AS Quantity_sold,
(b.Stock - COALESCE(SUM(o.Quantity), 0)) AS Available_Quantity
FROM books b
LEFT JOIN orders o
ON b.Book_ID = o.Book_ID
GROUP BY b.Book_ID , b.Stock;
```

## BUSINESS INSIGHTS

Data-derived observations from the supplied Books, Customers and Orders CSV files

### 1. Revenue and order value

The dataset contains 500 order records with total recorded revenue of $75,628.66. Average recorded order value is $151.26. These figures establish a baseline for tracking revenue and average order value over time.

### 2. Units sold and demand

The orders record 2,697 units in total. Mystery is the highest-volume genre in the joined order data, with 504 units recorded. This genre is a candidate for closer stock and merchandising review.

### 3. Inventory monitoring

The lowest listed stock in the Books table is 0 units for “Networked systemic implementation” (Book_ID 44). A low-stock report combined with order quantities can help flag titles for replenishment; current stock should be checked against actual demand before ordering.

### 4. Genre pricing

The average listed price of Fantasy books is $25.98. Comparing genre-level average prices with units sold and revenue can help evaluate whether a genre performs through volume, price, or both.

### 5. Product popularity

The book appearing in the greatest number of order records is “Implemented encompassing conglomeration”. Order frequency and quantity sold are different measures, so both should be reviewed before making a bestseller or replenishment decision.

### 6. Customer and geographic analysis

The most represented customer country in the supplied Customers table is Cuba (7 customers). Join customer locations to order values to compare markets, but avoid treating customer counts alone as evidence of profitability.